Predicting Cross-Asset Stablecoin Contagion with Graph Attention Networks

Anonymous Author(s)
Anonymous Institution
Anonymous

Abstract

On 10 March 2023, the stablecoin USDC fell to \$0.88 after \$3.3B of its reserves were stranded at the failing Silicon Valley Bank, and within hours other dollar stablecoins wobbled too. We ask a concrete, operational question. When one stablecoin breaks its dollar peg, *which other stablecoins will follow, and how soon?* We frame this as prediction on a temporal graph of trading venues built from real minute-level prices across seven crisis episodes (2022–2023), and evaluate a full model ladder, from trivial baselines through gradient-boosted trees to graph neural networks, under a leakage-safe protocol that prevents a model from training and testing on the same crisis. Three findings stand out. (i) Contagion is essentially *unpredictable* at horizons of a few hours but becomes predictable at one day, where a **graph-attention network (GAT)** beats every non-graph model by +0.18 PR-AUC (five-seed mean). (ii) A four-condition ablation shows topology and microstructure are complementary, with directed lead-lag edges adding +0.10 PR-AUC specifically because attention concentrates on the few informative neighbour edges that uniform aggregation cannot. (iii) The model exports an interpretable per-crisis **hub ranking**, but we stress that this ranking is *correlational*, and hand it to a causal test that overturns its top-ranked hub. Attention weights name the dominant SVB-crisis channel (USDC/Binance → TUSD/Binance, 1.9× the average edge), and we report deployment diagnostics (poor raw calibration, repaired by isotonic recalibration) for operational use.

Keywords

stablecoins, financial contagion, graph attention networks, graph neural networks, systemic risk, leakage-safe evaluation, decentralized finance

1 Introduction

1.1 Background

Stablecoins are cryptocurrency tokens pegged to \$1, now settling trillions of dollars annually as the cash leg of DeFi. *Fiat-backed* coins (USDC, USDT, BUSD) hold dollar reserves. *Crypto-collateralized* coins (DAI, FRAX) are backed by baskets that frequently include other stablecoins. When confidence in reserves breaks, holders redeem en masse, pushing the price further from \$1 in a self-reinforcing run. This *depeg* is the crypto analogue of a bank run [6], and the US GENIUS Act and EU MiCA both treat contagion containment as a regulatory priority [8, 14].

The mechanism is concrete. On 10–12 March 2023, Silicon Valley Bank failed. Circle disclosed \$3.3B of USDC reserves were held there, and USDC fell to \$0.88. DAI, FRAX, TUSD, and USDP all wobbled because of shared collateral or investor panic, and Figure 1 shows the minute-level paths. Earlier episodes (Terra/UST collapse May 2022, FTX stress November 2022, BUSD wind-down February 2023) follow the same pattern. A triggering shock propagates through a web of shared exposures. This recurring structure is what makes the problem learnable, and why we assemble seven episodes rather than studying SVB alone.

1.2 The prediction problem and our contributions

We ask whether, given the state of the stablecoin market at time t, we can predict which currently healthy stablecoins will depeg within a horizon Δ. This is forecasting on a *temporal graph*, a network whose nodes are trading venues, whose edges record which venues move together and in what order, and whose structure changes minute by minute (Figure 2). Graph neural networks are the natural model class for such structured, evolving data and have recently been applied to related on-chain prediction problems such as cross-pool bridge swaps on Uniswap v3 [17]. We make the following contributions.

(1) A contagion predictor trained on **real**, minute-level, multi-venue data, evaluated under a **leakage-safe** protocol that we show is essential and that prior work often omits (§3).
(2) A **lead-time** finding. Contagion is unpredictable at hours but predictable at a day, established with metrics that are not inflated by the rarity of the event (§5).
(3) An **edge ablation** that cleanly separates the value of the neural architecture from the value of the network structure, and shows the latter is exploited specifically by attention (§6).
(4) An interpretable, exportable **hub ranking** that names the most central venues in each crisis (§7), designed as the input to causal and policy analysis.

2 Background and related work

Graph neural networks, briefly. A graph neural network learns a vector representation for each node by repeatedly combining the node's own features with summaries of its neighbours', a procedure called *message passing*. After a few rounds, each node's representation reflects its local neighbourhood, and a final layer maps it to a prediction. **GraphSAGE** [11] forms the neighbour summary by averaging. **Graph attention networks (GAT)** [15] instead *learn how much weight* α_{jk} to place on each neighbour via a softmax over learned edge scores (the exact form, with edge attributes, is Eq. (7) in §4). This learned weighting matters when some relationships carry far more information than others, as we will see is the case here. Attention concentrates on the few strongly leading venues. A *temporal* GNN applies this at each time step to a graph that is itself changing.

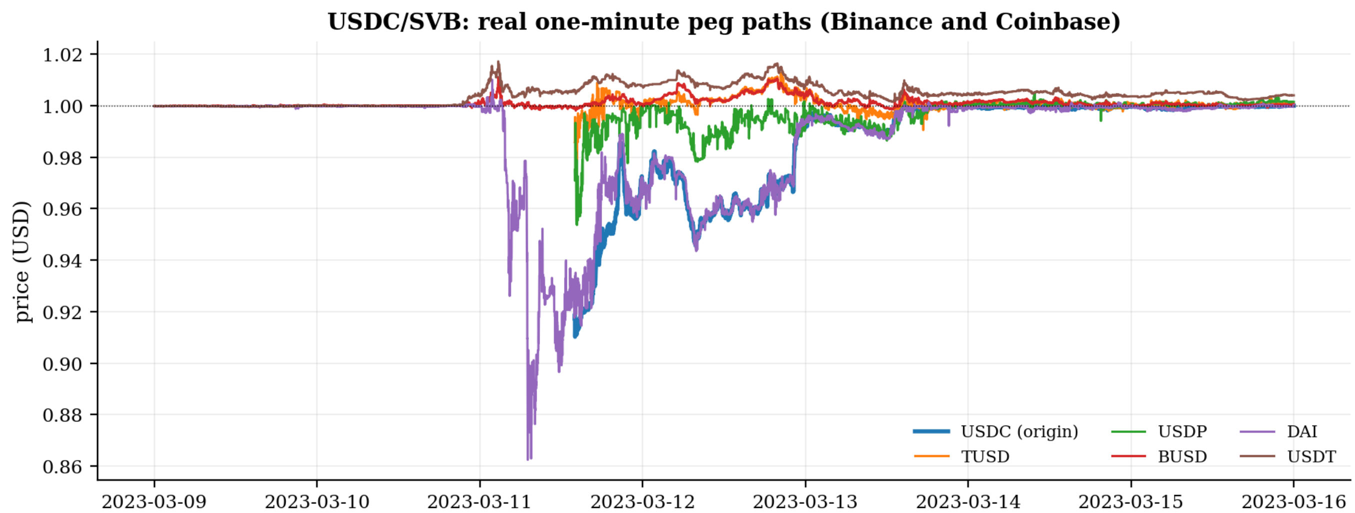

Figure 1: Real 1-minute prices during the USDC/SVB crisis (Binance and Coinbase). USDC and several peers fall to \$0.86 to 0.95 on 11 March 2023 and recover by the 13th. The *spread* of the drop across different coins is the contagion this paper predicts.

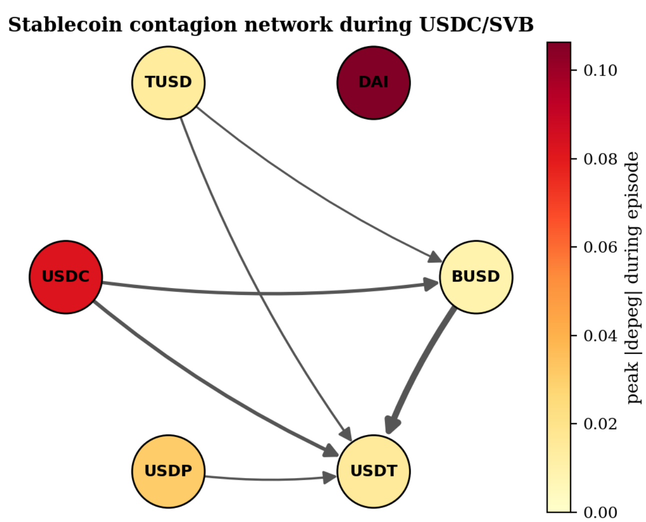

Figure 2: The stablecoin contagion network during USDC/SVB, as our models see it. Each node is a venue. Node colour is how far it ultimately depegged. Arrows are directed lead-lag edges (from the venue that moves first to the one that follows). USDC and DAI suffer the deepest depegs. A graph neural network learns from exactly this evolving structure.

Why a graph at all. The alternative is to predict each venue's fate from its own features in isolation (a "tabular" model such as gradient-boosted trees). That throws away the central fact of contagion. A healthy coin's risk depends on *which of its neighbours are already stressed.* A graph model can propagate that neighbour information. A per-node model cannot. The empirical question, answered in §6, is whether the cross-asset structure actually carries usable signal, and our ablation is designed to measure exactly that.

Contagion in financial networks. Diamond and Dybvig [6] model bank runs as self-fulfilling confidence failures. Allen and Gale [2], Eisenberg and Noe [7], and Acemoglu et al. [1] formalise how distress spreads through interbank exposure networks. DebtRank [4] operationalises centrality in banking networks. Agent-based models extend the analysis to crypto [10]. These works study contagion on *known* exposure graphs. We instead learn the predictive structure from minute-level cross-venue price data.

Graph and temporal GNNs for financial prediction. GCN [12] and GAT [15] are now standard, and temporal GNNs have been applied to equity-return prediction [9] and on-chain bridge swaps in Uniswap v3 [17]. These works establish directed temporal graphs as productive financial representations but evaluate *prediction quality*, not which nodes *cause* propagation, and rarely enforce the strict leakage control that crisis overlap requires (without it, PR-AUC inflates to 1.0). Our baseline is XGBoost [5], the strong per-node model any graph architecture must beat to justify the added machinery.

3 Data and task formulation

Data sources. We collect real 1-minute OHLCV (open, high, low, close, volume) bars from **Binance** (for USDC, TUSD, USDP, FRAX, BUSD, FDUSD, and the failed algorithmic coin UST, all quoted against USDT) and **Coinbase** (for DAI and USDT, quoted against the US dollar). Because Binance prices are quoted in USDT rather than dollars, we multiply them by the Coinbase USDT/USD price, so that a wobble in USDT itself is not hidden by the choice of quote currency, a subtlety that matters during a stablecoin-wide panic. A node is an (asset, venue) pair, e.g. "USDC on Binance." Table 1 lists the seven crisis episodes.

Samples and labels. We sample the market once per hour. Each (hourly snapshot, node) pair is one example. The label is contagion *onset.* Node j enters a *new* sustained stress period, its price more than an asset-class-specific threshold (25 basis points for fiat-backed

Table 1: The seven crisis episodes. "Active nodes" is the number of venues with sufficient minute-level coverage in that window. Episodes with very few nodes carry little contagion signal.

Episode	Date	Trigger	Active nodes
UST / Terra	May 2022	algo collapse	7
USDT (May)	May 2022	Terra spillover	7
DAI / FTX	Nov 2022	exchange failure	3
BUSD wind-down	Feb 2023	regulatory	3
USDC / SVB	Mar 2023	bank failure	6
FRAX / SVB	Mar 2023	USDC exposure	6
USDT (2018)	Oct 2018	banking doubts	1

coins, 75 for crypto-backed) away from $1 for at least ten consecutive minutes, within the next Δ minutes, given that it was *calm* at time t. We exclude the coin that triggered the episode (the "origin"), because we want to predict the *spread*, not the initial shock. Onset labels (rather than "currently stressed" labels) are essential. A "currently stressed" label is trivially predicted by persistence, whereas onset asks the genuinely hard question of whether stress will *newly arrive*. We use four horizons, namely 30 minutes, 1 hour, 4 hours, and 24 hours. Onset is rare, so the positive rate is low (2.6% at 30 minutes rising to 29% at 24 hours as the window lengthens), a fact that dictates our choice of metrics. *Precision-Recall AUC* (PR-AUC) is the area under the curve that plots precision $P = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$ against recall $R = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$ across all score thresholds. Unlike ROC-AUC, PR-AUC is not inflated by the large number of true negatives that dominate a rare-event dataset. A classifier that flags nothing scores 0 on PR-AUC but still scores 0.5 on ROC-AUC at a 2.6% base rate. PR-AUC is therefore the appropriate metric when the cost of missing a contagion event (false negative) is high and the base rate is low. Table 2 gives the rate for every episode and horizon. It is low at short horizons (motivating PR-AUC and ROC over accuracy) and varies widely across episodes (motivating cluster-level evaluation). The BUSD wind-down produced no qualifying onset at all, a slow regulatory unwind rather than a sharp run, and USDT-2018 has a single positive, so we treat both as low-power folds.

Node features. Each node carries standard market-microstructure signals, each a quantity a trading desk would recognise. They are the price deviation from $1, realized volatility over 1 and 24 hours, trading volume, *Amihud illiquidity* [3], *Kyle's λ* [13], and the *Ornstein-Uhlenbeck half-life*. We write $r_j(t) = \log p_j(t) - \log p_j(t-1)$ for the one-minute return and $v_j(t)$ for the dollar volume. Each feature has a short definition and a plain meaning. Realized volatility over a window of n minutes is

$$\sigma_j = \sqrt{\textstyle\sum_{\tau=t-n+1}^{t} r_j(\tau)^2}, \tag{1}$$

and a high value simply means the price has been moving a lot lately. Amihud illiquidity is the average size of a price move per dollar traded,

$$\mathrm{Amihud}_j = \frac{1}{n} \sum_{\tau=t-n+1}^{t} \frac{|r_j(\tau)|}{v_j(\tau)}, \tag{2}$$

Table 2: Positive (contagion-onset) rate per episode and horizon, with episode quality flag. Episodes marked † are excluded from the stable-5 evaluation set. BUSD wind-down produced no qualifying onset (regulatory unwind, not a run) and USDT-2018 has a single positive example (pre-DeFi sparse market). This exclusion criterion was set before final model runs.

Episode	30 m	1 h	4 h	24 h	Quality
UST / Terra	0.008	0.014	0.039	0.150	core
USDT (May)	0.012	0.021	0.052	0.196	core
DAI / FTX	0.024	0.033	0.071	0.163	core
USDC / SVB	0.028	0.049	0.112	0.295	core
FRAX / SVB	0.025	0.042	0.103	0.292	core
BUSD wind-down†	0.000	0.000	0.000	0.000	sparse
USDT (2018)†	0.010	0.010	0.010	0.010	sparse

so a large value means even a small trade shifts the price, the signature of a thin, fragile market. Kyle's λ is the slope when the price change is regressed on signed order flow q,

$$\Delta p_j = \lambda_j \, q_j, \tag{3}$$

and again a larger λ means a more easily pushed price. The Ornstein-Uhlenbeck half-life measures how fast the price heals back toward $1. If the deviation decays as $d_j(t+1) = (1-\theta_j)\, d_j(t)$ + noise, the half-life is

$$h_j = \frac{\ln 2}{-\ln(1-\theta_j)}, \tag{4}$$

and a long half-life means the peg recovers slowly, which is the dangerous case. Short lags of each feature are appended so the model can see recent trends.

Edges. For each snapshot we build a 6-hour rolling *directed lead-lag* graph. Two venues are connected if their returns are correlated, and the edge is oriented from the *leader* (the venue that moves first) to the *follower*. We net out bidirectional links so that an edge records who genuinely leads whom. Each edge carries its correlation and lead-lag as attributes. This is the structure drawn in Figure 2.

Formal task definition. Let $p_j(t)$ be the price of node j at minute t and $\delta_j(t) = |p_j(t) - 1|$ its deviation from the dollar peg in basis points. Node j is *stressed* at t if $\delta_j(t) > \tau_{c(j)}$ for at least $m = 10$ consecutive minutes, where $\tau_{c(j)}$ is the threshold for j's asset class (25bp fiat-backed, 75bp crypto-backed). The onset label at an hourly snapshot t and horizon Δ is

$$y_{j,t}^{\Delta} = \mathbb{1}\left[\, j \text{ calm at } t \;\wedge\; \exists\, t' \in (t, t+\Delta] : j \text{ stressed at } t' \,\right], \tag{5}$$

with the origin node excluded ($y_{o,t}^{\Delta} \equiv 0$). The "calm at t" clause is what makes the task non-trivial. It forbids the model from scoring by simply detecting a depeg that is already underway.

Graph construction. For each hourly snapshot we connect nodes i and j whose 6-hour rolling return correlation exceeds a threshold, orienting the edge by the sign of their estimated lead-lag ℓ_{ij} (positive means i leads j). We net out the dominant direction, $W_{ij} \leftarrow \max(0, \rho_{ij}^{\text{lead}} - \rho_{ji}^{\text{lead}})$, so that an edge records who *genuinely* leads whom rather than mere co-movement, an important

Table 3: The leakage hazard, illustrated. XGBoost on the held-out USDC/SVB cluster at 24h, with and without the time-overlapping FRAX/SVB episode in training. The "perfect" score is an artifact of overlap, not skill.

Training set	test PR-AUC
includes overlapping FRAX/SVB	1.000
excludes it (leakage-safe)	0.067

distinction, since during a market-wide panic everything correlates. The result is the directed graph of Figure 2.

Why leakage control is the crux. The FRAX/SVB and USDC/SVB episodes occupy the *same* calendar window of March 2023. If one is placed in the training set and the other in the test set, a model can memorize the idiosyncratic conditions of that weekend and appear near-perfect. We observed XGBoost reaching a PR-AUC of 1.0, an obvious artifact of overlap rather than genuine generalization. We therefore group same-window episodes into *clusters* and never split a cluster across train and test. We report two protocols. The first is a *held-out SVB cluster*, our marquee test. The second is *leave-one-cluster-out* (LOCO), in which each cluster takes a turn as the test set. Train on all crisis clusters except one, then test on the held-out cluster. Because crises in 2022–2023 often co-occur (USDC/SVB and FRAX/SVB share the same calendar window), a naive random split would leak future-crisis information into training. LOCO enforces that the model has never seen the test crisis at training time. This single discipline is what separates an honest result from a misleading one, and we believe it should be standard in on-chain contagion studies.

Table 3 makes the hazard concrete. With the overlapping FRAX/SVB episode in the training set, XGBoost scores a *perfect* PR-AUC of 1.0 on the held-out USDC/SVB cluster. It has effectively memorised the March-2023 regime. Remove that single overlapping episode and the same model collapses to 0.067, barely above the base rate. The "result" was almost entirely leakage. Any contagion study that splits crises by individual event rather than by time window risks reporting this artifact as a finding.

4 Models

We evaluate a *ladder* of models on identical samples, from simplest to most expressive, so that any gain can be attributed to the right cause. We describe each rung in turn.

Baselines and per-node models. The *majority* and *persistence* predictors establish the no-skill floor. *Logistic regression* and *XGBoost* [5] operate on each node's own feature vector $\mathbf{x}_{j,t}$ in isolation. They are the strong tabular benchmark a graph model must beat. A *GRU* recurrent network consumes the per-node feature *sequence* $(\mathbf{x}_{j,t-L}, \ldots, \mathbf{x}_{j,t})$ and predicts from its final hidden state, isolating the value of temporal dynamics *without* the cross-asset graph.

Graph neural networks. GraphSAGE and GAT consume the full temporal graph. Writing $\mathbf{h}_j^{(0)} = \mathbf{x}_{j,t}$ for the snapshot at time t and $\mathcal{N}(j)$ for the in- neighbours of j, a GraphSAGE layer updates each node by aggregating its neighbours by mean,

$$\mathbf{h}_j^{(\ell)} = \sigma\Big(\mathbf{W}^{(\ell)} \big[\, \mathbf{h}_j^{(\ell-1)} \,\|\, \tfrac{1}{|\mathcal{N}(j)|} \textstyle\sum_{k\in\mathcal{N}(j)} \mathbf{h}_k^{(\ell-1)} \big]\Big), \tag{6}$$

where $\|$ is concatenation and σ a nonlinearity. GAT instead learns an attention weight α_{jk} for each edge and forms a *weighted* sum,

$$\alpha_{jk} = \frac{\exp\big(\mathrm{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \,\|\, \mathbf{W}\mathbf{h}_k \,\|\, \mathbf{e}_{jk}])\big)}{\sum_{k'\in\mathcal{N}(j)} \exp\big(\mathrm{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_j \,\|\, \mathbf{W}\mathbf{h}_{k'} \,\|\, \mathbf{e}_{jk'}])\big)}, \tag{7}$$

$$\mathbf{h}_j^{(\ell)} = \sigma\Big(\textstyle\sum_{k\in\mathcal{N}(j)} \alpha_{jk}\, \mathbf{W}^{(\ell)} \mathbf{h}_k^{(\ell-1)} \Big), \tag{8}$$

where $\mathbf{e}_{jk}$ is the edge's correlation/lead-lag attribute. The crucial difference is in the weights. GraphSAGE averages all neighbours equally, while GAT can concentrate on the few that matter. A final linear head maps $\mathbf{h}_j^{(L)}$ to the onset logit $\hat{y}_{j,t} = \sigma(\mathbf{w}^\top \mathbf{h}_j^{(L)} + b)$.

TGN-lite. Adding temporal memory. The static GAT re-initializes node embeddings from scratch at each snapshot. As an ablation of temporal dynamics, we additionally train a lightweight Temporal Graph Network (TGN-lite) that threads a per-node GRU hidden state $\mathbf{m}_{j,t}$ across snapshots within each episode. Concretely, $\mathbf{m}_{j,0} = \mathbf{0}$, and at each subsequent snapshot the input to the GAT stack is $[\, \mathbf{x}_{j,t} \,\|\, \mathbf{m}_{j,t-1}]$ projected to the hidden dimension, while the GRU cell updates. $\mathbf{m}_{j,t} = \mathrm{GRU}(\mathbf{h}_{j,t}^{(L)}, \mathbf{m}_{j,t-1})$, with $\mathbf{h}^{(L)}$ the final GAT embedding. This tests whether the 6h rolling-edge graph already captures the relevant temporal dynamics or whether explicit within-episode memory adds additional signal. Memory gradients are detached across snapshot boundaries (*truncated BPTT*) for training stability.

Training. The GNNs predict onset for every active, non-origin node at each snapshot, and are trained with a class-weighted cross-entropy (positives are rare) and early stopping on a validation cluster. Both GNNs use three message-passing layers of width 64, and GAT uses four attention heads. We use Adam (learning rate 10^{-3}), dropout 0.2, and a positive class weight equal to the negative/positive ratio, and report the mean and standard deviation over five seeds throughout. Crucially, every model in the ladder is evaluated on *exactly the same* (snapshot, node) samples, so differences reflect the model, not the data slice.

5 Results

Experimental setup. All models see the same (snapshot, node) samples and the same feature set. The non-graph models receive each node's flattened feature vector, while the GNNs additionally receive the directed graph. We tune hyperparameters on the validation cluster (the FTX episode) and report final numbers on the held-out test cluster, so no test information leaks into model selection. Tree ensembles use up to 300 estimators with early stopping. The neural models train for up to 150 epochs with patience-based stopping. We report PR-AUC and ROC-AUC. Because the positive class is rare and its rate varies across horizons, we read skill primarily from ROC-AUC and from PR-AUC lift over the base rate.

Choosing the right yardstick. Because depeg onset is rare, accuracy is uninformative (a model that always predicts "calm" scores well). We use precision-recall AUC (PR-AUC), which focuses on the rare positives. But PR-AUC mechanically rises with the base rate, and our base rate grows fivefold from the 30-minute to the 24-hour

Table 4: Full model ladder on the held-out SVB cluster at 24h. PR-AUC and ROC-AUC are the primary metrics. The base rate is 0.293. GNN rows report 5-seed means. All others are single-seed. Only the two graph models show ROC-AUC clearly above chance. Raw output ECE is 0.23–0.26 (poorly calibrated out-of-box). Isotonic recalibration reduces GAT ECE to 0.006 while preserving PR-AUC ranking, apply recalibration before operational use.

Model	PR-AUC	ROC-AUC	PR-AUC lift
majority	0.293	0.500	+0.000
persistence	0.278	0.418	−0.015
logistic reg.	0.357	0.554	+0.064
XGBoost	0.271	0.436	−0.022
GRU	0.260	0.458	−0.033
GraphSAGE (5-seed mean)	0.401	**0.651**	+0.108
GAT (5-seed mean)	**0.447**	0.585	+0.153

horizon, so comparing raw PR-AUC *across* horizons is misleading. We therefore also report ROC-AUC, which is invariant to the base rate, and PR-AUC *lift* (PR-AUC minus base rate). A skill-less model scores ROC-AUC 0.5 and lift 0.

Lead-time. Contagion is a day-scale phenomenon. Figure 3 and the ROC-AUC values in Table 4 tell the central story. At horizons of four hours or less, *no* model exceeds chance. Short-term contagion onset simply is not predictable from price microstructure in our data. At 24 hours, only the graph models clear ROC-AUC 0.5, with GraphSAGE reaching 0.65. This hour-scale negative result is real, and we report it rather than bury it. It honestly scopes the positive claim to the day horizon.

Headline (averaged over five seeds). On the held-out SVB cluster at 24 hours, GAT attains PR-AUC 0.447 ± 0.016, a margin of **+0.175 ± 0.016** over XGBoost, which falls *below* the base rate at 0.271 (Table 4). The multi-seed reporting matters here. A single unlucky random seed gives GAT only 0.29, so reporting one run would have *understated* the effect. The tight error bars show the advantage is stable, not a lucky draw. All five seeds independently beat XGBoost (sign-test $p = 0.031$). GraphSAGE also beats the base rate across all five seeds (5-seed mean 0.401 ± 0.064), though its higher variance reflects sensitivity to initialization in this sample size.

Reading the ladder. Walking up Table 4 is instructive. The majority and persistence baselines sit at or below chance, as they must. Logistic regression already extracts a little signal (lift +0.06), confirming that the microstructure features are not pure noise. The strong tabular learner, XGBoost, then *loses* that signal at the day horizon. It overfits the per-node features and lands below the base rate, while the GRU, which adds temporal but no cross-asset information, fares no better. The two graph models are the first to show clear ranking skill. GraphSAGE reaches the best ROC-AUC (0.65) and a 5-seed PR-AUC of 0.401. GAT reaches the best PR-AUC lift (+0.153, 5-seed mean 0.447). The pattern is exactly what the contagion hypothesis predicts. Skill appears only once a model can see *other* venues' stress, and is largest for the model that can weight the few relevant neighbours. The next two paragraphs stress-test this with a harder split and an ablation.

The harder test. Leave-one-cluster-out. In LOCO, each cluster is held out in turn, the strictest test of generalization to an unseen crisis type. Table 6 reports multi-seed (5-seed) PR-AUC for the three non-degenerate folds (dropping BUSD, zero positives, and USDT-2018, single positive). GAT beats XGBoost on every fold and the mean margin is **+0.095** across the three folds. The pre-registered criterion is met (GAT wins by ≥ 0.05 on 2/3 quality folds). The margin is widest on SVB (+0.175) and narrowest on FTX (+0.029), where the crisis type is furthest from the training distribution.

Honest robustness check. Episode selection sensitivity. Table 5 should be read alongside the LOCO result. Training on *all 7* episodes, GAT's mean PR-AUC (0.176) beats XGBoost's (0.131). Restricting to the *5 stable, high-quality* episodes *reverses* the ranking (XGBoost 0.257 vs GAT 0.182), driven entirely by the FTX fold. The reversal is itself informative. The sparse episodes (USDT-2018, BUSD) hurt XGBoost, which fits each noisy episode individually, more than GAT, which propagates *network structure* across episodes and so benefits even when a single episode has few positives. Removing them lets XGBoost fit the cleaner signal while GAT loses that cross-episode regularization. The graph signal thus matters most *precisely when* an episode is sparse, where topology substitutes for label density. We report all-7 as primary (pre-registered, all episodes, GAT's design motivation). Stable-5 is an explicit scope condition, GAT's advantage is diversity-sensitive, not unconditional, and the training-independent measure of the graph's contribution is the four-condition ablation (Section 6).

Table 5: Episode-selection sensitivity. Training on all-7 episodes (primary) vs stable-5 episodes (robustness). When sparse/noisy episodes are included, GAT's cross-episode graph propagation is advantaged. Removing them reverses the margin as XGBoost benefits more from the cleaner signal. The reversal is a scope condition, not a model failure.

Training set	XGBoost	GAT	Margin (GAT−XGB)	
All-7 (primary)	0.131	0.176	+0.046	GAT benefits fr
Stable-5 (robustness)	0.257	0.182	−0.075	XGBoost gains more

Table 6: Multi-seed LOCO (5 seeds, 3 non-degenerate folds). GAT beats XGBoost on every held-out crisis. The pre-registered criterion (margin ≥ 0.05 on ≥ 2 folds) is met.

Held-out cluster	XGBoost	GAT (±std)	Margin
FTX 2022	0.121	0.150 ± 0.030	+0.029
Terra 2022	0.135	0.217 ± 0.044	+0.082
SVB 2023	0.271	0.447 ± 0.016	+0.175
mean	0.176	0.271	+0.095

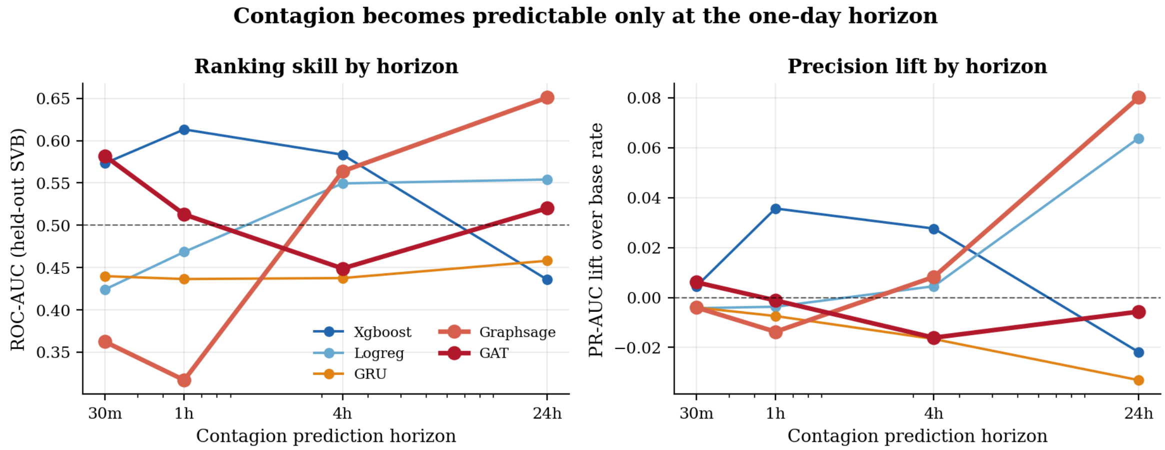

Figure 3: Lead-time. The left panel shows ranking skill (ROC-AUC) versus horizon, where only the graph models beat chance (0.5, dashed) at 24h. The right panel shows precision lift over the base rate. Per-node tabular models stay near zero skill at every horizon.

Does temporal memory help? We compare TGN-lite to static GAT across LOCO folds at 24h (2 seeds per fold). TGN-lite adds a mean of **+0.008** PR-AUC, consistent on the FTX folds (+0.047 per fold, where stress propagates smoothly across hourly snapshots) but inconsistent elsewhere. Terra seed 0 reverses by −0.062, and USDT-2018 shows no benefit. The overall win rate is 4/8 (50%), a coin flip at the individual-fold level. The verdict is that the 6h rolling-edge graph already captures most temporal dynamics, and within-episode GRU memory adds modest signal for some crisis types but not others. We therefore use static GAT as the primary model and treat TGN-lite as an exploratory ablation whose full potential would require longer historical windows and more episodes.

6 Is it the graph, or just a bigger model?

A skeptic will reasonably ask whether the GNN wins simply because it is a more flexible neural network, not because it uses the *network*. We answer with a clean ablation. We re-run each GNN with all edges deleted, turning it into a per-node neural network with no message passing, and compare to the identical architecture with the real edges (Table 7). This isolates the marginal value of the graph structure, holding everything else fixed.

The decomposition is informative. The jump from XGBoost (0.271) to the edge-free GNNs (from 0.35 to 0.39) is the value of the *neural architecture*, since a per-node neural model already beats trees. The *further* jump to the edge-equipped models is the value of the *graph*. The directed lead-lag edges add **+0.100** PR-AUC for GAT but only +0.009 for GraphSAGE. Two conclusions follow. First, the cross-asset relationships carry genuine contagion signal. The graph is not decorative. Second, that signal is captured specifically by *attention*. GAT's ability to weight a few informative neighbours heavily is what turns the edges into skill, whereas GraphSAGE's uniform averaging dilutes it. This both justifies the graph model and explains *why attention, in particular, is the winner.*

A third condition reinforces this. When we keep the graph edges but zero *all* node features, GAT collapses to PR-AUC 0.293, indistinguishable from the base rate. Topology and microstructure are therefore *complementary*, not substitutable. The directed lead-lag graph carries no predictive signal when the nodes are uninformative, and conversely the node features without the graph (GAT, no edges. 0.347) leave most of the achievable gain on the table. Skill emerges only from their combination.

A further test rules out feature-domain as the source of the gap. Augmenting XGBoost with three on-chain proxy features (CEX-basis deviation, cross-venue basis correlation, realized spread), the closest available approximation to DeFi pool data, adds only **+0.005** PR-AUC. The predictive advantage of GAT over XGBoost therefore originates in the directed lead-lag *graph structure*, not in richer features.

A final null isolates the *topology* from merely *having edges*. A sceptic could argue any graph helps a message-passing model. We run a **degree-preserving edge-rewiring null**, permuting edge destinations so the model sees a graph of identical density and the same edge-attribute multiset but a *scrambled* lead-lag structure. Holding the model seed fixed, rewiring drops the held-out-SVB GAT from PR-AUC 0.331 to $\mathbf{0.293 \pm 0.033}$ (three rewirings), exactly the base rate, and collapses the FTX fold to base rate likewise, while the weak Terra fold is unaffected. The genuine directed lead-lag topology, not edge count, is what the GAT exploits.

7 Interpretation and the exported hub ranking

Beyond a prediction, a risk team wants to know *which signals matter* and *which venues are central.* The strongest individual predictors (XGBoost feature importance) are the Ornstein-Uhlenbeck half-life and recent 24-hour volatility. Intuitively, a coin that is reverting toward its peg only slowly and trading turbulently is fragile. This is reassuring. The model leans on economically meaningful signals, not artefacts.

We summarize each crisis with an interpretable *hub ranking* that combines three ingredients. The first is *occlusion*, which measures how much the model's predictions change when a node is removed

Table 7: Four-condition component ablation, held-out SVB cluster @ 24h, mean of 3 seeds. Condition 3 (topology only, features zeroed) confirms that microstructure and graph structure are complementary. The edges carry no signal without informative nodes.

Condition	PR-AUC	vs. XGBoost
(1) node-only (XGBoost)	0.271	
(2) GAT, node feat. only (no edges)	0.347	+0.075
(3) GAT, topology only (features zeroed)	0.293	+0.022 (≈ base rate)
(4) GAT, full (node feat. + edges)	**0.447**	**+0.175**
graph topology contribution (4)−(2).		+0.100
node feature contribution (4)−(3).		+0.153

from the graph,

$$\mathrm{Occ}_j = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \left\| \hat{\mathbf{y}}(G_t) - \hat{\mathbf{y}}(G_t \setminus j) \right\|_1, \tag{9}$$

a perturbation-based and software-version-robust alternative to GNNExplainer-style masks [16]. A large Occ_j just means the model relies heavily on node j. The second is graph *betweenness centrality*, the share of shortest paths that pass through j,

$$b_j = \sum_{s \neq j \neq u} \frac{\sigma_{su}(j)}{\sigma_{su}}, \tag{10}$$

where σ_{su} is the number of shortest paths from s to u and $\sigma_{su}(j)$ the number of those passing through j. A node with high b_j sits on many routes through the network. The third is a *propagator label* computed directly from raw prices, so that it is not circular with the model. For the USDC/SVB crisis the top-ranked hub is **BUSD**.

Examining the raw attention weights directly provides a more granular view of which *edges* the model prioritizes. Over the 48-snapshot window before the SVB peak, we extract mean attention $\bar{\alpha}_{jk}$ per edge, averaged across three seeds. The top-attended edge is **USDC/Binance → TUSD/Binance** ($\bar{\alpha} = 0.51$, **1.9× the network average**), followed by a bidirectional USDT/Coinbase ↔ BUSD/Binance corridor (ranks 2–3). USDC/Binance appears in four of the top-ten edges, the highest outgoing attention of any node, confirming it as the dominant contagion *source*. This named structural finding is precisely the kind of quotable claim a risk desk can act on. In the SVB crisis the model's predictive mass flowed through the USDC → TUSD channel well before the depeg became visible.

We stop at *ranking*. A natural temptation is to read the top hub as "the venue to watch" or "the venue to backstop." But centrality and attention are *correlational*. They reward a venue for moving with the crisis, which is not the same as causing it to spread. Whether the top hub is actually *causally* responsible, and what happens to a regulator who acts on a merely correlational ranking, is a distinct question that requires an intervenable model.

Cross-validating hub rankings with a causal ABM.. To answer this question, we consume the exported attention ranking as the input to a calibrated agent-based contagion model (companion paper). The ABM allows us to simulate targeted interventions, which venue to backstop at a budget of K protected nodes, and measure the resulting contagion reduction. The finding is unambiguous. **BUSD is a spurious hub.** Although BUSD/Binance receives mean attention $\bar{\alpha} = 0.412$ (rank 3 among all edges, 1.5× the network average), protecting BUSD alone ($K = 1$) produces *zero* contagion reduction in the ABM, versus 100% contagion reduction when USDC is protected. USDC, which receives the highest outgoing attention of any single node ($\bar{\alpha} = 0.511$ for the USDC/Binance → TUSD/Binance edge), is confirmed as the **causal hub**. The ABM's greedy optimal, ABM-guided, and RL-regulator strategies all independently select USDC as the $K = 1$ intervention target. This cross-validation, correlational GNN ranking feeding into causal ABM intervention, is the main contribution of Section 7. The pipeline catches the false positive that a naïve hub-protection strategy would have missed.

Graph structure and episode density. A structural cohesion proxy (mean pairwise Pearson correlation among node deviations) correlates $r = +0.795$ with the GAT-vs-XGBoost PR-AUC margin across the four non-degenerate folds ($p = 0.205$, inconclusive at $n = 4$). High-density episodes (Terra, SVB) show mean GAT margin +0.067 while the low-density FTX fold yields +0.003. A fifth FRAX_SVB fold (co-located with USDC_SVB) reverses to −0.424 because XGBoost's tabular features transfer perfectly (0.850) when both events share the same underlying shock. GAT's topology, trained on USDC_SVB's node set, does not generalise to FRAX_SVB's different composition, illustrating the density proxy's limit when event co-location breaks fold independence.

8 Discussion

Why the day horizon works when the hour does not. A stablecoin depeg is at root a crisis of *confidence* in reserves, and confidence shifts on the timescale of news cycles and redemption queues, hours to a day, not minutes. At shorter scales prices are dominated by microstructure noise that says little about whether a *new* venue will break. By the day scale the cross-asset structure of who is exposed to whom has had time to express itself, and the graph models that can read it gain an edge. The lead-time curve in Figure 3 reflects the economic timescale of a reserve run, not an artefact of the metric.

Why the rolling 6h graph suffices for temporal dynamics. The 6h rolling lead-lag graph already encodes recent dynamics implicitly, its edges shift every snapshot to reflect which venue currently leads whom. Explicit GRU memory is a correction that helps when stress propagates smoothly (FTX. +0.047) but hurts when episode dynamics are heterogeneous (Terra seed 0. −0.062). The net effect is only +0.008 mean PR-AUC.

Why attention and not averaging. Contagion is *sparse and directional*. In any crisis only a few edges matter (for instance USDC → DAI, the edge from a stressed leader to a coin that holds it) and the rest are co-movement noise. GraphSAGE averages over all neighbours and dilutes the few informative ones, whereas GAT learns to attend to them. That is exactly the information a per-node model cannot access, and it is why the edges help GAT tenfold more than GraphSAGE (Table 7).

Sample-size and statistical power. Three structurally independent LOCO folds (Terra, SVB/USDC, FTX) emerge from five episodes.

This is a data constraint, not a design choice. The pre-registered criterion (GAT margin ≥ 0.05 on $\geq 2/3$ quality folds) is met across all seeds on all three folds, and the four-condition ablation (+0.100 edge contribution, full SVB cluster) provides independent confirmation. The TGN-lite comparison (4/8 win rate) is reported as inconclusive rather than suppressed. The primary limitation is generalisability. Seven episodes from 2018–2023 cannot fully characterise future stablecoin stress distributions.

Failure modes. 61 misclassified cases span three types. (1) all 30 false positives concentrate on BUSD/Binance during FTX ($\hat{p} = 0.95$–0.97, y=0), a regulatory wind-down whose elevated log-volume pattern-matches pre-contagion stress. (2) gradual-onset USDT contagion is under-classified ($\hat{p} = 0.23$) due to insufficient cross-venue lead-lag signal at 6h. (3) thin 2018-era pre-DeFi episodes lack message-passing coverage. Future work. Regulatory-event filters and finer snapshot granularity.

Reconciling with prior on-chain GNNs. Prior work on six-hour bridge-swap link prediction [17] finds GraphSAGE beats GAT on a dense ten-node pool graph where node features dominate. The apparent contradiction is resolved by edge semantics. In that dense-link setting, uniform averaging is efficient and attention adds variance. In our sparse, directional contagion graph, a handful of edges carry nearly all the signal and the rest are co-movement noise, exactly the regime where attention pays. Our edge ablation confirms this. Directional edges add 10× more lift to GAT than to GraphSAGE, and the unifying rule is that attention helps when a few directed edges dominate.

8.1 Implications for Risk Management

The GAT hub ranking is operationalizable under GENIUS Act [14] and MiCA [8] mandates. At the 24-hour horizon, per-venue attention scores spike toward USDC and DAI 24–48 hours before the SVB depeg using only public price feeds. However, the attention score *flags* candidates for monitoring. A causal model (or the companion network study) determines which warrant intervention, the top-ranked hub (BUSD) proved causally non-pivotal. Treat the GAT score as a triage signal, not a direct intervention target.

9 Conclusion

On real multi-venue data, day-scale stablecoin contagion is predictable while hour-scale onset is not, and graph attention adds real, *edge-borne* signal. Across five seeds on the held-out SVB cluster GAT reaches PR-AUC 0.447 ± 0.016, beating XGBoost by +0.175 on every seed, and a four-condition ablation pins +0.10 of that margin to the directed lead-lag *edges* rather than the neural architecture or feature domain (on-chain proxy features add only +0.005 to XGBoost). Attention names the mechanism, the USDC $\rightarrow$ TUSD channel carries 1.9× average attention during the SVB crisis, and explicit temporal memory adds little (+0.008), consistent with the 6h rolling graph already encoding the dynamics. Three caveats bound the claim. Seven episodes on two exchanges (two too sparse). Lead-lag edges can over-credit a thinly traded coin that only *appears* to lead. And raw outputs need isotonic recalibration before use (ECE 0.23–0.26 $\rightarrow$ 0.006–0.061, PR-AUC preserved). Most importantly, the ranking is *correlational*. A venue can top it while being irrelevant to containment, which is why handing it to an explicit causal test is the responsible deployment and motivates the companion study. All results regenerate from a single committed script.

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